Full-Stack Web Development with Agentic AI: A Design Thinking Approach

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Beginner | 12 weeks (48 study days) | 12 Weeks

Course Overview

A 12-week comprehensive learning plan for beginner final-year engineering students to develop a full-stack web application integrated with agentic AI capabilities, following the Design Thinking Playbook methodology. The plan culminates in a portfolio-ready working software product demonstrating intelligent autonomous behavior.

Level
Beginner

Duration
12 weeks (48 study days)

Total Weeks
12

Prerequisites

What you should know

- Basic computer literacy
- Familiarity with logical thinking and problem-solving
- Motivation to learn programming and software development
- Access to a computer with internet
- Understanding of basic math concepts

Target Audience

Perfect for these learners

- Final-year engineering students with limited software development experience
- Individuals aspiring to build full-stack applications
- Learners interested in integrating AI into web solutions
- Teams looking to apply Design Thinking to software projects
- Beginner developers seeking practical project experience

Course Curriculum

Weekly breakdown designed to guide your learning

1 Week 1

4 days 8 topics

Monday

Introduction to Design Thinking & Project Kick-off

Study Time: 09:30 AM - 12:30 PM

Practice Exercises

- Understand the 5 stages of Design Thinking (Empathize, Define, Ideate, Prototype, Test).
- Review project objectives and final deliverables.
- Form project teams (if applicable) and brainstorm initial problem areas for the application.

Fundamentals of Web Development: HTML Basics

Study Time: 01:00 PM - 03:30 PM

Practice Exercises

- Learn HTML document structure (tags, attributes, elements).
- Create a simple static webpage with headings, paragraphs, and lists.
- Resource: MDN Web Docs (HTML Basics)

Tuesday

Design Thinking: Empathize Phase - User Research

Study Time: 09:30 AM - 12:30 PM

Practice Exercises

- Learn about basic user interview techniques and observation methods.
- Conduct mock interviews with peers or potential users (based on brainstormed ideas).
- Start collecting initial insights on user needs and pain points related to the project theme.

Fundamentals of Web Development: CSS Basics

Study Time: 01:00 PM - 03:30 PM

Practice Exercises

- Learn CSS selectors, properties, and values for styling.
- Apply CSS to the previously created HTML page (e.g., colors, fonts, basic layouts).
- Resource: W3Schools CSS Tutorial

Wednesday

Design Thinking: Empathize Phase - Persona & Empathy Mapping

Study Time: 09:30 AM - 12:30 PM

Practice Exercises

- Create foundational user personas based on initial research findings.

- Develop empathy maps to understand user feelings, thoughts, and motivations.
- Identify key user challenges and opportunities for a software solution.

Responsive Design with Flexbox/Grid (Intro)

Study Time: 01:00 PM - 03:30 PM

Practice Exercises

- Understand the concept of responsive web design for different screen sizes.
- Implement a simple responsive layout using CSS Flexbox or Grid.
- Practice adjusting layouts for mobile and desktop views.

Thursday

Milestone 1: Empathize Phase Completion & Documentation

Study Time: 09:30 AM - 12:30 PM

Practice Exercises

- Consolidate all user research findings, personas, and empathy maps into a brief report.
- Prepare a brief presentation or summary on the identified user problem space.
- Conduct a peer review and feedback session on empathy work.

Version Control with Git and GitHub Basics

Study Time: 01:00 PM - 03:30 PM

Practice Exercises

- Install Git and set up a GitHub account.
- Learn basic Git commands (init, add, commit, push, pull, clone).
- Create a new repository for the project and push initial web files.

Week 2

2 4 days 8 topics

3 **Week 3**
4 days 8 topics

4 **Week 4**
4 days 8 topics

5 **Week 5**
4 days 8 topics

6 **Week 6**
4 days 8 topics

7 **Week 7**
4 days 8 topics

8 **Week 8**
4 days 8 topics

9 **Week 9**
4 days 8 topics

10 **Week 10**
4 days 8 topics

11 **Week 11**
4 days 7 topics

12 **Week 12**
4 days 5 topics

Ready to Begin Your Journey?

Join thousands of learners who have successfully completed this course and advanced their skills.

Begin Learning

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